

NM_007294.4(BRCA1):c.68_69del (p.Glu...

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Reviewed by expert panel

★★★★

Pathogenic

for [BRCA1-related cancer predisposition](#)

Classification is based on the expert panel submission
Jun 2024 by ClinGen ENIGMA BRCA1 and BRCA2 Variant Curation Expert Pane...
FDA RECOGNIZED DATABASE

Somatic

No data submitted for somatic clinical impact

Somatic

No data submitted for oncogenicity

Variant Details

Variant details

Identifiers:

NM_007294.4(BRCA1):c.68_69del (p.Glu23fs)

Type and length:

Microsatellite, 2 bp

Location:

Variation ID: 17662 Accession: VCV000017662.180

Cytogenetic:

17q21.31 17: 43124028-43124029 (GRCh38) [[NCBI](#) [UCSC](#)] 17: 41276045-41276046 (GRCh37) [[NCBI](#) [UCSC](#)]

Timeline in ClinVar:

HGVS:

First in ClinVar	Last submission	Last evaluated
Germline	Apr 12, 2013	May 9, 2026
		Jun 11, 2024

Nucleotide	Protein	Molecular consequence
NM_007294.4:c.68_69del MANE SELECT	NP_009225.1:p.Glu23fs	frameshift
NM_007294.4:c.68_69delAG MANE SELECT		frameshift
NM_001407571.1:c.-123_-122AG[1]		

Protein change:

E23fs

Other names:

... more HGVS

NP_009225.1:p.Glu23ValfsTer17 Canonical SPDI: NC_000017.11:43124027:CTCT:CT

NM_007294.4(BRCA1):c.68_69del

c.185delAG

p.Glu23fs*17

187delAG

185delAG

Links : [Breast Cancer Information Core \(BIC\) \(BRCA1\): 185&base_change=del AG](#)

[ClinGen: CA003783](#)

[Genetic Testing Registry \(GTR\): GTR000558872](#)

[Genetic Testing Registry \(GTR\): GTR000566692](#)

[OMIM: 113705.0003](#)

[dbSNP: rs80357914](#)

[VarSome](#)

Global minor allele frequency (GMAF) :

-

-

Genes

Gene	OMIM	ClinGen Gene Dosage Sensitivity Curation		Variation Viewer	Related variants	
		HI score	TS score		Within gene	All
BRCA1		Sufficient evidence for dosage pathogenicity	No evidence available	GRCh38 GRCh37	13907	15892

Conditions - Germline

Condition	Classification (# of submissions)	Review status	Last evaluated	Variation/condition record
Breast-ovarian cancer, familial, susceptibility to, 1	Pathogenic (35)	★★★★	Feb 11, 2026	RCV000019230.73
Pancreatic cancer, susceptibility to, 4	risk factor (1)	★★★★	Oct 1, 2008	RCV000019231.12

click to load more conditions

Condition	Classification (# of submissions)	Review status	Last evaluated	Variation/condition record
Hereditary breast ovarian cancer syndrome	Pathogenic (14)	★★★★	Jan 28, 2026	RCV000034761.55
Familial cancer of breast	Pathogenic (4)	★★★★	Apr 2, 2024	RCV000056295.22
Hereditary cancer-predisposing syndrome	Pathogenic (6)	★★★★	Mar 31, 2025	RCV000131394.27
not provided	Pathogenic (16)	★★★★	Feb 1, 2026	RCV000213650.84
Breast and/or ovarian cancer	Pathogenic (3)	★★★★	Mar 4, 2022	RCV000735481.13
Ovarian neoplasm	Pathogenic (1)	★★★★	Dec 1, 2018	RCV000785197.10
Malignant tumor of breast	Pathogenic (1)	★★★★	-	RCV001353942.9
Invasive medullary breast carcinoma	Pathogenic (1)	★★★★	-	RCV001836711.10

click to load more conditions

Submissions - Germline

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows
Pathogenic (Jun 11, 2024)  Contributing to aggregate classification	★★★★ (CSpec BRCA1/2ACMG Rules Specifications V1.0)	BRCA1-related cancer predisposition (Autosomal dominant inheritance)	ClinGen ENIGMA BRCA1 and BRCA2 Variant Curation Expert Panel, ClinGen FDA RECOGNIZED DATABASE Accession: SCV004101425.2 First in ClinVar: Nov 11, 2023 Last updated: Jun 17, 2024	

Other databases

<https://erepo.clinicalgenome.org/evrepo/ui/interpretation/8cc9bd83-9b2d-4a38-ba40-c2010955350d>

Comment:

The c.68_69del variant in BRCA1 is a deletion of two nucleotides, predicted to encode a frameshift with consequent premature termination of the protein at codon 17 of the frameshift, or amino acid 39 (p.Glu23ValfsTer17). This variant is present in gnomAD v2.1 (exomes only, non-cancer subset) or gnomAD v3.1 (non-cancer subset) but is below the ENIGMA BRCA1/2 VCEP threshold >0.00002 for BS1_Supporting (PM2_Supporting, BS1, and BA1 are not met). Frameshift variant predicted to cause a premature stop codon in biologically-relevant-exon 3 leading to nonsense mediated decay (PVS1 met). The ENIGMA BRCA1/2 VCEP considered multiple lines of functional and clinical evidence to define exon-specific weights for PTC in BRCA1, and results indicate that strong evidence towards pathogenicity may be applied for a PTC variant in BRCA1 exon 3 (PM5_Strong (PTC)). Reported by one calibrated study to exhibit protein function similar to pathogenic control variants (PMID: 32546644) (PS3 met). In summary, this variant meets the criteria to be classified as a Pathogenic variant variant for BRCA1-related cancer predisposition based on the ACMG/AMP criteria applied as specified by the ENIGMA BRCA1/2 VCEP (PVS1, PM5_Strong (PTC), PS3). [\(less\)](#)

Observation 1

Collection method: curation
Allele origin: germline
Affected status: unknown

Pathogenic (Nov 20, 2015)  Not contributing to aggregate classification	★★★★ (Shirts et al. (Genet Med 2016))	Breast-ovarian cancer, familial, susceptibility to, 1	University of Washington Department of Laboratory Medicine, University of Washington Accession: SCV000266036.1 First in ClinVar: Mar 20, 2016 Last updated: Mar 20, 2016	
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Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes
Number of individuals with the variant: 1
Clinical Features:
pancreatic cancer (present)
Age: 60-69 years

Observation 2

Collection method: clinical testing
Allele origin: germline
Affected status: no
Number of individuals with the variant: 1

Pathogenic (Jul 01, 2015)  Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Department of Medical Genetics, Oslo University Hospital Accession: SCV000564332.1 First in ClinVar: Dec 06, 2016 Last updated: Dec 06, 2016	
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Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 13					
Pathogenic (Jan 01, 2017) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Genologica Medica Additional submitter: Servicio Andaluz de Salud, Hospital Universitario Virgen de la Victoria Accession: SCV000577914.1 First in ClinVar: Dec 06, 2016 Last updated: Dec 06, 2016		
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes Test name: BRCAsafe Family history: yes Ethnicity/Population group: Causasians Geographic origin: Spain Tissue: Blood Secondary finding: no Platform type: Sanger sequencing					
Pathogenic (Apr 18, 2017) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Hereditary breast ovarian cancer syndrome	Cancer Genetics and Genomics Laboratory, British Columbia Cancer Agency Study: The Canadian Open Genetics Repository (COGR) Accession: SCV000586860.1 First in ClinVar: Apr 16, 2017 Last updated: Apr 16, 2017	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: unknown	
Pathogenic (Apr 20, 2017) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Hereditary breast ovarian cancer syndrome	Department of Pathology and Molecular Medicine, Queen's University Study: The Canadian Open Genetics Repository (COGR) Accession: SCV000588022.1 First in ClinVar: Apr 16, 2017 Last updated: Apr 16, 2017	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: unknown	
Pathogenic (Dec 02, 2016) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	{Breast-ovarian cancer, familial, 1}	Genetic Services Laboratory, University of Chicago Accession: SCV000593688.1 First in ClinVar: Dec 06, 2016 Last updated: Dec 06, 2016	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: yes	
Pathogenic (Aug 10, 2017) N Not contributing to aggregate classification	★★★★ (LabCorp Variant Classification Summary - May 2015)	Hereditary breast ovarian cancer syndrome	Women's Health and Genetics/Laboratory Corporation of America, LabCorp Accession: SCV000699291.1 First in ClinVar: Dec 26, 2017 Last updated: Dec 26, 2017		
Publications:					
PubMed: 12566964, 20924075					
Comment:					
Variant summary: The BRCA1 c.68_69delAG (p.Glu23Valfs) variant results in a premature termination codon, predicted to cause a truncated or absent BRCA1 protein due to nonsense mediated decay, which are commonly known mechanisms for disease. The variant of interest is one of the most common Jewish founder mutations. Multiple publications have cited the variant in affected individuals. This variant was found in 29/120972 control chromosomes at a frequency of 0.0002397, which does not exceed the estimated maximal expected allele frequency of a pathogenic BRCA1 variant (0.0010005). In addition, multiple clinical diagnostic laboratories/reputable databases classified this variant as pathogenic. Taken together, this variant is classified as pathogenic. (less)					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: unknown					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Pathogenic (Jan 22, 2019) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	HudsonAlpha Institute for Biotechnology, HudsonAlpha Institute for Biotechnology Study: AGHL_GT Accession: SCV000993552.1 First in ClinVar: Oct 01, 2019 Last updated: Oct 01, 2019		

Observation 1

Collection method: research
Allele origin: unknown
Affected status: unknown
Number of individuals with the variant: 1

Pathogenic (Jan 01, 2020) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Hereditary Breast Carcinoma	GeneKor MSA Accession: SCV000693502.3 First in ClinVar: Feb 24, 2015 Last updated: May 04, 2020		
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Comment:

This sequence change deletes 2 nucleotides from exon 2 of the BRCA1 mRNA (c.68_69delAG), causing a frameshift after codon 23 and the creation of a premature translation stop signal 17 amino acid residues later - p.(Glu23Valfs*17). This is expected to result in an absent or disrupted protein product. This variant, also known as BRCA1_185delAG or 187delAG, is one of three main pathogenic founder variants in the Ashkenazi Jewish population although it has been observed in individuals from other ethnicities as well (PMID: 23199084). This mutation has been described in the mutation database ClinVar (Variation ID: 17662). ([less](#))

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes

Pathogenic (Jun 05, 2014) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	not provided	Clinical Genetics and Genomics, Karolinska University Hospital Accession: SCV001449768.1 First in ClinVar: Dec 12, 2020 Last updated: Dec 12, 2020		
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Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes
Number of individuals with the variant: 15

Pathogenic (Dec 15, 2020) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Familial cancer of breast	Department of Pediatrics, Memorial Sloan Kettering Cancer Center Accession: SCV001478110.1 First in ClinVar: Jun 19, 2021 Last updated: Jun 19, 2021		
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Publications:

PubMed: [28873162](#)

Observation 1

Collection method: research
Allele origin: germline
Affected status: no

Pathogenic (Sep 01, 2020) N Not contributing to aggregate classification	★ ★ ★ ★ (LMM Criteria)	Hereditary breast ovarian cancer syndrome (Autosomal dominant inheritance)	Laboratory for Molecular Medicine, Mass General Brigham Personalized Medicine Accession: SCV000271311.5 First in ClinVar: May 29, 2016 Last updated: May 29, 2021		
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Publications:

PubMed: [9042909](#), [9145676](#)

Comment:

The p.Glu23ValfsX17 variant in BRCA1 has been reported in numerous individuals with hereditary breast and ovarian cancer (HBOC) and is a known pathogenic Ashkenazi Jewish founder variant (Struwing 1997 PMID: 9145676, Abeliovich 2013 PMID: 9042909). It has also been identified in 0.4% (42/10368) of Ashkenazi Jewish and 0.009% (11/128780) of European chromosomes by gnomAD (<http://gnomad.broadinstitute.org>). Please note that this frequency is low enough to be consistent with the frequency of HBOC in the general population. This frameshift variant is predicted to alter the protein's amino acid sequence beginning at position 23 and lead to a premature termination

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
codon 17 amino acids downstream. This alteration is then predicted to lead to a truncated or absent protein. Heterozygous loss of function of the BRCA1 gene is an established disease mechanism in individuals with HBOC. Additionally, this variant was classified as Pathogenic on April 22, 2016 by the ClinGen-approved ENIGMA expert panel (ClinVar SCV000282348.1). In summary, this variant meets criteria to be classified as pathogenic for HBOC in an autosomal dominant manner based upon the predicted impact to the protein and presence in multiple affected individuals. ACMG/AMP Criteria applied: PVS1, PS4_strong. (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown Number of individuals with the variant: 19					
Pathogenic (-) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Molecular Endocrinology Laboratory, Christian Medical College Accession: SCV002003976.1 First in ClinVar: Oct 30, 2021 Last updated: Oct 30, 2021		
Publications: PubMed: 23788959 Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes					
Pathogenic (Oct 07, 2021) N Not contributing to aggregate classification	★★★★ (Sema4 Curation Guidelines)	Hereditary cancer-predisposing syndrome	Sema4, Sema4 Accession: SCV002537890.1 First in ClinVar: Jun 24, 2022 Last updated: Jun 24, 2022 Comment: The BRCA1 c.68_69delAG (p.E23VfsX17) is a well characterized pathogenic variant that has been associated with autosomal dominant hereditary breast and ovarian cancer syndrome (PMID: 9042909). ... (more)		
Publications: PubMed: 29446198, 9042909, 9150153 Observation 1 Collection method: curation Allele origin: germline Affected status: unknown					
Pathogenic (Jul 02, 2018) N Not contributing to aggregate classification	★★★★ (Mendelics Assertion Criteria 2017)	Hereditary breast and ovarian cancer syndrome	Mendelics Accession: SCV000839317.2 First in ClinVar: Oct 10, 2018 Last updated: Dec 11, 2022	Observation: 1 - Collection method: clinical testing - Allele origin: unknown - Affected status: unknown	
Pathogenic (Mar 22, 2022) N Not contributing to aggregate classification	★★★★ (St. Jude Assertion Criteria 2020)	Hereditary breast ovarian cancer syndrome	St. Jude Molecular Pathology, St. Jude Children's Research Hospital Accession: SCV002525990.2 First in ClinVar: Jun 18, 2022 Last updated: Dec 24, 2022		
Comment: The BRCA1 c.68_69del (p.Glu23ValfsTer17) change causes a frameshift and the creation of a premature stop codon. This change is predicted to cause protein truncation or absence of the protein due to nonsense mediated decay (PVS1). This variant has a maximum subpopulation frequency of 0.41% in gnomAD v2.1.1, where it is primarily found in the Ashkenazi Jewish population (https://gnomad.broadinstitute.org/variant/17-41276044-ACT-A?dataset=gnomad_r2_1). This variant has been reported in multiple individuals with breast cancer and/or ovarian cancer (PMID: 14576434, 26718727, 21503673, 22430266, 23633455, 22752604), prostate (PMID: 22516946) and pancreatic cancer (PMID: 20711688, 24737347, 23658460, 26440929). It is a well-established pathogenic founder variant in the Ashkenazi Jewish population (PMID: 9042909, 22430266). This alteration is also known as 185delAG or 187delAG in the literature. In summary, this variant meets criteria to be classified as pathogenic based on the ACMG/AMP criteria: PVS1, PS4. (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown					
Pathogenic (Apr 20, 2023)	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Genomic Medicine Lab, University of California San Francisco	Observation: 1	

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
N Not contributing to aggregate classification			Accession: SCV004847144.1 First in ClinVar: Apr 20, 2024 Last updated: Apr 20, 2024	• Collection method: clinical testing • Allele origin: maternal • Affected status: no	
Pathogenic (Jan 16, 2024)	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1 Pancreatic cancer, susceptibility to, 4 Fanconi anemia, complementation group S	Fulgent Genetics, Fulgent Genetics Accession: SCV002814217.2 First in ClinVar: Dec 31, 2022 Last updated: Jan 25, 2025	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: unknown	
N Not contributing to aggregate classification		Explanation for multiple conditions: Uncertain. The variant was classified for several related diseases, possibly a spectrum of disease; the variant may be associated with one or more the diseases.			
Pathogenic (Mar 25, 2025)	★ ★ ★ ★ (Hauer et al. (Genet Med. 2018))	not provided (Unknown mechanism)	Institute of Human Genetics, FAU Erlangen, Friedrich-Alexander-Universität Erlangen-Nürnberg Accession: SCV005900507.1 First in ClinVar: Mar 29, 2025 Last updated: Mar 29, 2025		
N Not contributing to aggregate classification					
Comment: This variant has been identified by standard clinical testing. Female patient with metastasised triple negative breast cancer Selected ACMG criteria: Pathogenic (I).PP5;PM2;PVS1 (less)					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1					
Pathogenic (-)	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	GreenArray Genomic Research & Solutions of Accurate Diagnostic Private Limited Accession: SCV002097608.2 First in ClinVar: Feb 20, 2022 Last updated: Apr 13, 2025		
N Not contributing to aggregate classification					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: no Sex: female					
Pathogenic (-)	★ ★ ★ ★ (ACMG Guidelines, 2015)	Invasive medullary breast carcinoma	GreenArray Genomic Research & Solutions of Accurate Diagnostic Private Limited Accession: SCV002097613.2 First in ClinVar: Feb 20, 2022 Last updated: Apr 13, 2025		
N Not contributing to aggregate classification					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes Sex: female					
Pathogenic (Feb 19, 2025)	★ ★ ★ ★ (Ambry Variant Classification Scheme 2023)	Hereditary cancer-predisposing syndrome	Ambry Genetics Accession: SCV000186370.10 First in ClinVar: Aug 06, 2014 Last updated: Apr 28, 2025		
N Not contributing to aggregate classification					

Publications:

PubMed: 10090881, 10464624, 15994883, 16168118, 21990299, 22006311, 22430266, 22588877, 23401782, 24737347, 24830819, 26440929, 26543556, 26718727, 27062684, 27283172, 27489289, 27553291, 27767231, 27819678, 27989354, 28324225, 28390335, 28477318, 28526081, 28528518, 28655807, 28993434, 29161300,

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
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[29446198](#), [29470806](#), [29560538](#), [29752822](#), [30067863](#), [30093976](#), [30430080](#), [30555256](#), [30883245](#), [31013702](#), [31029168](#), [31076742](#), [31159747](#), [31209999](#), [31372034](#), [31454914](#), [32075053](#), [33471991](#), [7550349](#), [7611277](#)

Comment:

The c.68_69delAG pathogenic mutation, located in coding exon 1 of the BRCA1 gene, results from a deletion of two nucleotides at nucleotide positions 68 to 69, causing a translational frameshift with a predicted alternate stop codon (p.E23Vfs*17). This alteration is one of three well-characterized Ashkenazi Jewish founder mutations, with an overall carrier frequency of nearly 1% in this population (Struewing JP et al. Nat. Genet. 1995 Oct;11:198-200; Schubert EL et al. Genet Test, 1997;1:41-6; Hartge P et al. Am. J. Hum. Genet. 1999 Apr;64:963-70). This germline mutation has been reported in individuals of Ashkenazi Jewish descent, as well as in cohorts of other ethnicities, with hereditary breast and ovarian cancer (HBOC) syndrome, including individuals with male breast cancer, pancreatic cancer, and prostate cancer (Antoniou AC et al. J. Med. Genet. 2005 Jul;42:602-3; Lucas AL et al. Clin. Cancer Res. 2013 Jul;19:3396-403; Lucas AL et al. Cancer. 2014 Jul;120:1960-7; Bernards SS et al. Gynecol Oncol. 2016 Feb;140:221-5; Azzollini J et al. Eur J Intern Med. 2016 Jul;32:65-71; Na R et al. Eur Urol. 2017 05;71:740-747; Gabaldonacute; Barrios X et al. Fam. Cancer. 2017 Oct;16:477-489; Alemar B et al. PLoS One. 2017 Nov;12:e0187630; Chan GHJ et al. Oncotarget. 2018 Jul;9:30649-30660; Schayek H et al. Breast Cancer Res Treat. 2018 Jul;170:399-404; Brand R et al. Cancer. 2018 09;124:3520-3527; Singh J et al. Breast Cancer Res Treat. 2018 Jul;170:189-196; Wen WX et al. J Med Genet. 2018 02;55:97-103; Cock-Rada AM et al. Fam Cancer. 2018 01;17:23-30; Mehta A et al. Cancer Manag Res. 2018 Nov;10:6505-6516; Li JY et al. Int J Cancer. 2019 01;144:281-289; Abe T et al. J Clin Oncol. 2019 05;37:1070-1080; Ashour M et al. Cancer Manag Res. 2019 Jul;11:6275-6284). In a large case control-study, this variant was reported in 30/60,466 breast cancer cases and in 8/53,461 controls (Dorling et al. N Engl J Med. 2021 02;384:428-439). Of note, this alteration is also designated as 185delAG, 187delAG, and 189delAG in published literature. In addition to the clinical data presented in the literature, this alteration is expected to result in loss of function by premature protein truncation or nonsense-mediated mRNA decay. As such, this alteration is interpreted as a disease-causing mutation. ([less](#))

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Pathogenic (Feb 06, 2025) N Not contributing to aggregate classification	★ ★ ★ ★ (ClinGen BRCA1BRCA2 ACMG Specifications BRCA1 V1.0.0)	Hereditary cancer-predisposing syndrome	Molecular Diagnostics Laboratory, Catalan Institute of Oncology Accession: SCV006060713.1 First in ClinVar: May 03, 2025 Last updated: May 03, 2025	
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Other databases

<https://spadahc.ciberiscii.es/variant/GRCH37/17%2041276045%20CT/->

Comment:

PVS1, PM5, PTC_Strong, PS3 c.68_69del, located in exon 2 of the BRCA1 gene, consists in the deletion of two nucleotides, causing a translational frameshift with a predicted alternate stop codon, p.(Glu23Valfs*17). This alteration is expected to result in loss of function by premature protein truncation and nonsense-mediated mRNA decay (PVS1, PM5, PTC_Strong). This variant is found in 52/236532 in the gnomAD v2.1.1 database, exome only non-cancer data set. The SpliceAI algorithm predicts no effect on splicing. Reported by one calibrated study to exhibit protein function similar to pathogenic control variants (PMID: 32546644) (PS3). In addition, it has been reported in ClinVar (75x as pathogenic), LOVD databases (254x as pathogenic, 1x pPAT, 3x not classified) and also classified as pathogenic reviewed by an expert panel (ENIGMA (22/04/2016)). Variant allele predicted to encode a truncated non-functional protein. The c.68_69del variant in BRCA1 is a deletion of two nucleotides, predicted to encode a frameshift with consequent premature termination of the protein at codon 17 of the frameshift, or amino acid 39 (p.Glu23ValfsTer17). This variant is present in gnomAD v2.1 (exomes only, non-cancer subset) or gnomAD v3.1 (non-cancer subset) but is below the ENIGMA BRCA1/2 VCEP threshold >0.00002 for BS1_Supporting (PM2_Supporting, BS1, and BA1 are not met). Frameshift variant predicted to cause a premature stop codon in biologically-relevant-exon 3 leading to nonsense mediated decay (PVS1 met). The ENIGMA BRCA1/2 VCEP considered multiple lines of functional and clinical evidence to define exon-specific weights for PTC in BRCA1, and results indicate that strong evidence towards pathogenicity may be applied for a PTC variant in BRCA1 exon 3 (PM5_Strong (PTC)). Reported by one calibrated study to exhibit protein function similar to pathogenic control variants (PMID: 32546644) (PS3 met). In summary, this variant meets the criteria to be classified as a Pathogenic variant for BRCA1-related cancer predisposition based on the ACMG/AMP criteria applied as specified by the ENIGMA BRCA1/2 VCEP (PVS1, PM5_Strong (PTC), PS3)). Based on currently available information, the variant c.68_69del is classified as a pathogenic variant according to ClinGen-BRCA1 and BRCA2 Guidelines version 1.0.0. ([less](#))

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Pathogenic (Mar 03, 2025) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1 (Autosomal dominant inheritance)	Neuberg Centre For Genomic Medicine, NCGM Accession: SCV004100646.4 First in ClinVar: Nov 04, 2023 Last updated: Sep 06, 2025	
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Comment:

This variant is also known as 185delAG or 187delAG. Functional in vitro assays showed that this variant was deleterious [Bouwman P, et al. 2013]. ([less](#))

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes

Pathogenic (Jan 28, 2026) N Not contributing to aggregate classification	★ ★ ★ ★ (Invitae Variant Classification Sherloc (09022015))	not provided	Revvity Omics, Revvity Accession: SCV002020211.4 First in ClinVar: Nov 29, 2021 Last updated: Sep 06, 2025	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: unknown
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Pathogenic (Jan 28, 2026) N Not contributing to aggregate classification	★ ★ ★ ★ (Invitae Variant Classification Sherloc (09022015))	Hereditary breast ovarian cancer syndrome	Labcorp Genetics (formerly Invitae), Labcorp Accession: SCV000077108.20 First in ClinVar: Jul 03, 2013 Last updated: Feb 15, 2026	
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Publications:

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
	PubMed: 15994883 , 20104584 , 22430266 , 23658460 , 24737347 , 28492532 , 8571953 , 8651293 , 9042909 , 9921907				
Comment: This sequence change creates a premature translational stop signal (p.Glu23Valfs*17) in the BRCA1 gene. It is expected to result in an absent or disrupted protein product. Loss-of-function variants in BRCA1 are known to be pathogenic (PMID: 20104584). This variant is present in population databases (rs386833395, gnomAD 0.4%), and has an allele count higher than expected for a pathogenic variant. This premature translational stop signal has been observed in individual(s) with breast and ovarian cancer, and pancreatic cancer (PMID: 9042909, 15994883, 22430266, 23658460, 24737347). It is commonly reported in individuals of Ashkenazi Jewish ancestry (PMID: 8571953, 8651293, 9042909, 9921907, 15994883, 22430266, 23658460, 24737347). This variant is also known as 185delAG or 187delAG. ClinVar contains an entry for this variant (Variation ID: 17662). For these reasons, this variant has been classified as Pathogenic. (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown					
Pathogenic (Mar 29, 2024)	★★★★ (ACMG Guidelines, 2015)	Hereditary cancer-predisposing syndrome	Institute for Genomic Medicine (IGM) Clinical Laboratory, Nationwide Children's Hospital Accession: SCV007449511.1 First in ClinVar: Feb 15, 2026 Last updated: Feb 15, 2026		
N Not contributing to aggregate classification Publications: PubMed: 32546644 Comment: This variant is predicted to result in loss of function through nonsense-mediated decay of the encoded transcript or premature truncation of the encoded protein in a gene in which loss of function is a known mechanism of disease (ACMG/AMP: PVS1). Well-established functional studies have demonstrated this variant to have a damaging effect on protein function or splicing (ACMG/AMP: PS3; PMID:32546644). This variant has been reported at an elevated frequency in affected individuals/in multiple affected individuals in the literature (ACMG/AMP: PS4). (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown					
Pathogenic (Feb 11, 2026)	★★★★ (St. Jude Assertion Criteria 2020)	Breast-ovarian cancer, familial, susceptibility to, 1	St. Jude Molecular Pathology, St. Jude Children's Research Hospital Accession: SCV007518309.1 First in ClinVar: Mar 14, 2026 Last updated: Mar 14, 2026		
N Not contributing to aggregate classification Comment: The BRCA1 c.68_69del p.(Glu23ValfsTer17) change deletes two nucleotides to cause a frameshift and the creation of a premature stop codon. This change is predicted to cause protein truncation or absence of protein due to nonsense-mediated decay. This variant, which is also known as 185delAG and 187delAG, has been reported in multiple individuals with breast and/or ovarian cancer (PMID: 14576434, 26718727, 21503673, 22430266, 23633455, 22752604), prostate cancer (PMID: 22516946) and pancreatic cancer (PMID: 20711688, 24737347, 23658460, 26440929). This variant is a well-established founder variant in the Ashkenazi Jewish population (PMID: 9042909, 22430266) and has a maximum subpopulation frequency of 0.41% in the Ashkenazi Jewish population in gnomAD v2.1.1 (https://gnomad.broadinstitute.org). In summary, this variant meets criteria to be classified as pathogenic. (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown					
Pathogenic (Feb 01, 2026)	★★★★ (CeGaT Center For Human Genetics Tuebingen Variant Classification Criteria Version 2)	not provided	CeGaT Center for Human Genetics Tuebingen Accession: SCV001250428.37 First in ClinVar: May 12, 2020 Last updated: Apr 25, 2026		
N Not contributing to aggregate classification Comment: BRCA1: PVS1, PM2, PS4:Moderate (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 15					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows
Pathogenic (Nov 03, 2014) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Michigan Medical Genetics Laboratories, University of Michigan Accession: SCV000195875.1 First in ClinVar: May 06, 2016 Last updated: May 06, 2016	
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes Tissue: Blood				
Pathogenic (Jun 14, 2016) N Not contributing to aggregate classification	★ ★ ★ ★ (ICSL Variant Classification 20161018)	Hereditary Breast and Ovarian Cancer	Illumina Laboratory Services, Illumina Accession: SCV000403078.2 First in ClinVar: Dec 06, 2016 Last updated: Dec 06, 2016	
Publications: PubMed: 15994883, 21643751, 22430266, 22763381, 23788959, 23867111, 7837387, 7894492, 8642955, 8651293, 9042909, 9145676, 9921907 Comment: The c.68_69delAG (p.Glu23ValfsTer17) variant, also commonly known as 185delAG, is a frameshift variant that is very well described in the literature as one of the most frequent variants found in breast cancer (Shattuck-Eidens et al. 1995; Offit et al. 1996). It was first identified in a study by Simard et al. (1994) in index cases from four unrelated Canadian families with hereditary breast and ovarian cancer (HBOC). Wang et al. (2012) conducted a meta-analysis of over 29 studies published between 2000 and 2010 and determined the overall frequency of the p.Glu23ValfsTer17 variant in 2128 breast cancer cases to be 0.072. The variant is one of three known common founder germline variants primarily found in individuals of Ashkenazi Jewish heritage (Struewing et al. 1997; Abellovich et al. 1997; Laitman et al. 2013) and has been detected in 20% of Ashkenazi Jewish women diagnosed with breast cancer before age 42 years (Offit et al 1996). In the Ashkenazi Jewish population, the variant is associated with a risk of between 56% and 83% of breast cancer and of between 14% and 58% risk of ovarian cancer by the age of 70 (Struewing et al. 1997; Antoniou et al. 2005; Finkelmann et al. 2012). The variant has been observed in individuals of other ethnicities (Wang et al. 2012; Laitman et al. 2013). The variant is reported at a frequency of 0.00042 in the European (Non-Finnish) population of the Exome Aggregation Consortium. Based on the available evidence, the p.Glu23ValfsTer17 variant is classified as pathogenic for hereditary breast and ovarian cancer. (less)				
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown				
Pathogenic (Feb 23, 2017) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Familial cancer of breast (Autosomal dominant inheritance)	Baylor Genetics Accession: SCV000540943.1 First in ClinVar: Feb 24, 2015 Last updated: Feb 24, 2015	
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1				
Pathogenic (Dec 03, 2017) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Genomic Research Center, Shahid Beheshti University of Medical Sciences Accession: SCV000746289.1 First in ClinVar: Apr 09, 2018 Last updated: Apr 09, 2018	
Observation 1 Collection method: clinical testing Allele origin: inherited Affected status: yes Geographic origin: Iran				
Pathogenic (May 23, 2018) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Hereditary cancer-predisposing syndrome	Institute for Biomarker Research, Medical Diagnostic Laboratories, L.L.C. Accession: SCV000803154.1 First in ClinVar: Oct 21, 2017 Last updated: Oct 21, 2017	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: unknown

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Pathogenic (May 25, 2017) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Human Genome Sequencing Center Clinical Lab, Baylor College of Medicine Accession: SCV000839891.1 First in ClinVar: Apr 09, 2018 Last updated: Apr 09, 2018		⬆
Comment: The c.68_69del (p.Glu23Valfs*17) variant in the BRCA1 gene has been detected multiple patients with breast cancer and/or ovarian cancer [PMID 14576434, 26718727, 21503673, 22430266, 23633455, 22752604, referred as c.185delAG in some publications]. This variant is a founder mutation in Ashkenazi Jews with a frequency of about 1% [PMID 14576434]. The variant has also been detected in patients with prostate [PMID 22516946] and pancreatic cancer [PMID 20711688, 24737347, 23658460, 26440929]. This one bp deletion in exon 2 results in a frameshift and the creation of a premature stop codon. Functional in vitro assays showed that this variant was deleterious [PMID 23867111]. This variant has been reported in 31 individuals from the ExAC database (http://exac.broadinstitute.org/variant/22-29091226-TA-T). This variant is thus classified as pathogenic (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown					
Pathogenic (Apr 02, 2020) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Department of Molecular Diagnostics, Institute of Oncology Ljubljana Accession: SCV001499755.1 First in ClinVar: Mar 07, 2021 Last updated: Mar 07, 2021	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: yes	
Pathogenic (Apr 19, 2022) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Hereditary breast ovarian cancer syndrome	National Health Laboratory Service, Universitas Academic Hospital and University of the Free State Accession: SCV002026049.2 First in ClinVar: Apr 23, 2022 Last updated: Apr 23, 2022		⬆
Publications: DOI: 10.3389/fgene.2022.834265 Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 3 Geographic origin: South Africa					
Pathogenic (Oct 02, 2015) N Not contributing to aggregate classification	★ ★ ★ ★ (CIMBA Mutation Classification guidelines May 2016)	Breast-ovarian cancer, familial, susceptibility to, 1	Consortium of Investigators of Modifiers of BRCA1/2 (CIMBA), c/o University of Cambridge Accession: SCV000326390.4 First in ClinVar: Nov 05, 2016 Last updated: Dec 11, 2022	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: unknown	
Pathogenic (Oct 16, 2019) N Not contributing to aggregate classification	★ ★ ★ ★ (GeneDx Variant Classification Process June 2021)	not provided	GeneDx Accession: SCV000278833.11 First in ClinVar: May 29, 2016 Last updated: Jun 10, 2023		⬆
Comment: Frameshift variant predicted to result in protein truncation or nonsense mediated decay in a gene for which loss-of-function is a known mechanism of disease; Truncating variants in this gene are considered pathogenic by a well-established clinical consortium and/or database; Also known as 185delAG or 187delAG; This variant is associated with the following publications: (PMID: 18594935, 26071757, 26221963, 27259235, 28049106, 26822237, 29176636, 31948886, 22703879, 22535016, 22516946, 22009639, 23633455, 23658460, 22763381, 23086583, 19906413, 21503673, 22430266, 20711688, 22752604, 22006311, 7611277, 23867111, 25556971, 24737347, 26440929, 26689913, 26641009, 27553291, 23788959, 26718727, 26681312, 27836010, 29321669, 27062684, 28528518, 11802208, 28477318, 29339979, 29752822, 28324225, 27989354, 29907814, 29431110, 29560538, 26556299, 10464624, 29470806, 28390335, 28993434, 26357657, 29161300, 30702160, 30067863, 30152102, 30630528, 30122538, 30186769, 30720243, 30322717, 30093976, 31159747, 30113427, 31372034, 30489631, 30612635, 31454914, 31528241, 29625052, 31447099, 31980526, 34308366, 33646313, 10739756, 10733239, 11597388, 9042909, 12220453, 31589614, 32341426, 9634504, 10885601, 32719484, 33556450, 12491828, 32338768, 30875412, 30787465, 16030426, 33224455) (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes					
Pathogenic (Mar 04, 2022) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast and/or ovarian cancer	CHEO Genetics Diagnostic Laboratory, Children's Hospital of Eastern Ontario Study: Canadian Open Genetics Repository Accession: SCV000901065.2 First in ClinVar: May 06, 2019 Last updated: Mar 11, 2023		⬆

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
	Publications: PubMed: 25741868				
	Observation 1				
	Collection method: clinical testing Allele origin: germline Affected status: unknown				

Pathogenic (Aug 09, 2022)	★ ★ ★ ★ (NYGC Assertion Criteria 2020)	Breast-ovarian cancer, familial, susceptibility to, 1	New York Genome Center Accession: SCV002764582.2 First in ClinVar: Dec 17, 2022 Last updated: May 20, 2023	
N Not contributing to aggregate classification				

Publications:
PubMed: [14576434](#), [22752604](#), [23633455](#)

Comment:
The c.68_69del variant in BRCA1 (also referred as c.185delAG) is an established pathogenic variant [ClinVar ID:17662] that is predicted (p.(Glu23fs)) to result in truncated or absent BRCA1 protein due to nonsense mediated decay. The c.68_69del variant is a founder mutation in Ashkenazi Jewish population with a frequency of about 1% [PMID: 14576434]. The c.68_69del variant has been reported in multiple individuals with breast and/or ovarian, prostate, and pancreatic cancers [PMID:14576434, 20711688, 22516946, 23633455, 22752604]. Based on available evidence, this c.68_69del (p.(Glu23fs)) variant identified in BRCA1 is reported as Pathogenic. ([less](#))

Observation 1
Collection method: clinical testing Allele origin: germline Affected status: unknown Clinical Features: Hyperlipidemia (present) , Diabetes mellitus (present) Test name: whole genome sequencing Zygosity: 1 Single Heterozygote Secondary finding: yes Platform name: NovaSeq 6000

Observation 2
Collection method: clinical testing Allele origin: germline Affected status: unknown Clinical Features: Hyperlipidemia (present) , Hepatic steatosis (present) Test name: whole genome sequencing Zygosity: 1 Single Heterozygote Secondary finding: yes Platform name: NovaSeq 6000

Observation 3
Collection method: clinical testing Allele origin: germline Affected status: unknown Clinical Features: Hyperlipidemia (present) , Hepatic steatosis (present) , Type 2 diabetes mellitus (present) Zygosity: 1 Single Heterozygote Secondary finding: yes Platform Name: NovaSeq 6000

Pathogenic (Nov 03, 2021)	★ ★ ★ ★ (ACMG Guidelines, 2015)	not provided	Institute for Clinical Genetics, University Hospital TU Dresden, University Hospital TU Dresden Accession: SCV002009407.3 First in ClinVar: Oct 30, 2021 Last updated: Jul 16, 2023	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: not provided
N Not contributing to aggregate classification				

Pathogenic (Mar 09, 2023)	★ ★ ★ ★ (Quest Diagnostics criteria)	not provided	Quest Diagnostics Nichols Institute San Juan Capistrano Accession: SCV000296272.7 First in ClinVar: Jun 24, 2016 Last updated: Jan 06, 2024	
N Not contributing to aggregate classification				

Publications:
PubMed: [10464624](#) [10733239](#) [10739756](#) [10885601](#)
[11179017](#) [11597388](#) [11802208](#) [11896106](#) [12220453](#)
[12491828](#) [14576434](#) [15024741](#) [15994883](#) [16030426](#)
[16168118](#) [16941470](#) [17688236](#) [18594935](#) [19147582](#)
[19329713](#) [19906413](#) [20104584](#) [20651350](#) [20711688](#)
[21203900](#) [21503673](#) [22006311](#) [22009639](#) [22274685](#)
[22430266](#) [22516946](#) [22535016](#) [22703879](#) [22752604](#)
[22763381](#) [23086583](#) [23199084](#) [23531862](#) [23633455](#)
[23658460](#) [23683081](#) [23788959](#) [23855721](#) [23867111](#)
[23941904](#) [24737347](#) [26295337](#) [26357657](#) [26440929](#)
[26556299](#) [26641009](#) [26681312](#) [26689913](#) [26718727](#)
[26822237](#) [27062684](#) [27259235](#) [27553291](#) [27836010](#)
[27989354](#) [28049106](#) [28324225](#) [28390335](#) [28477318](#)
[28528518](#) [28993434](#) [29161300](#) [29176636](#) [29321669](#)
[29339979](#) [29431110](#) [29446198](#) [29470806](#) [29560538](#)

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
			29625052 29752822 29907814 30067863 30093976 30113427 30122538 30152102 30186769 30322717 30430080 30489631 30630528 30702160 30720243 30787465 30875412 31159747 31372034 31447099 31528241 31589614 31948886 31980526 32338768 32341426 32719484 33224455 33556450 33646313 33754277 33758026 34308366 35039532 35264596 35356428 35377489 35459234 35710434 7611277 8533757 8764110 9042909 9145676 9634504		

Comment:

This frameshift variant alters the translational reading frame of the BRCA1 mRNA and causes the premature termination of BRCA1 protein synthesis. The frequency of this variant in the general population, 0.0041 (42/10368 chromosomes, <http://gnomad.broadinstitute.org>), is consistent with pathogenicity. In the published literature, the variant has been reported as one of the common pathogenic BRCA founder variants in the Ashkenazi Jewish population (PMIDs: 22430266 (2012), 14576434 (2003), 8841191 (1996), 7550349 (1995)). The variant has been reported in individuals from multiple populations with a personal or family history of breast and/or ovarian cancer (PMIDs: 35377489 (2022), 33758026 (2022), 35039532 (2022), 35264596 (2022), 35356428 (2022), 35710434 (2022), 33646313 (2021), 32341426 (2020), 31528241 (2019), 31372034 (2019), 31159747 (2019), 30875412 (2019), 30630528 (2019), 30489631 (2019)) and prostate cancer (PMIDs: 33556450 (2021), 32338768 (2020), 31948886 (2020)). Functional studies report the variant disrupts proper protein function (PMID: 35459234 (2022)). Based on the available information, this variant is classified as pathogenic. ([less](#))

Observation 1

Collection method: clinical testing
Allele origin: unknown
Affected status: unknown

Pathogenic (May 27, 2024)	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Department of Clinical Genetics, Copenhagen University Hospital, Rigshospitalet Accession: SCV005045898.1 First in ClinVar: Jun 02, 2024 Last updated: Jun 02, 2024	
N Not contributing to aggregate classification				

Comment:

PVS1; PM5_PTC_Strong ([less](#))

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: yes

Pathogenic (Mar 27, 2024)	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Baylor Genetics Accession: SCV004212668.2 First in ClinVar: Dec 30, 2023 Last updated: Jun 17, 2024	Observation: 1 • Collection method: clinical testing • Allele origin: unknown • Affected status: unknown
N Not contributing to aggregate classification				

Pathogenic (Jul 29, 2022)	★★★★ (ACMG Guidelines, 2015)	not provided	Clinical Genetics Laboratory, Skane University Hospital Lund Accession: SCV005198273.1 First in ClinVar: Aug 25, 2024 Last updated: Aug 25, 2024	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: yes
N Not contributing to aggregate classification				

Pathogenic (Sep 13, 2024)	★★★★ (ACMG Guidelines, 2015)	BRCA1-related cancer predisposition (Autosomal dominant inheritance)	All of Us Research Program, National Institutes of Health Accession: SCV004823702.2 First in ClinVar: Apr 20, 2024 Last updated: Dec 14, 2024 Comment: This study involves interpretation of variants in research participants for the purpose of population health screening. Participant phenotype was not available at the time of ... (more)	
N Not contributing to aggregate classification				

Publications:

PubMed: 15994883, 21643751, 22430266, 24312913,
30152102, 30480775, 33471991, 35020120, 7550349,
7611277, 7837387, 7894492, 8531968, 8533757, 8571953,
8642955, 8651293, 9042909, 9145676, 9150153

Comment:

This variant deletes 2 nucleotides in exon 2 of the BRCA1 gene, causing a frameshift and a premature translational stop signal. This variant is also known as 185delAG and 187delAG in the literature. This variant is expected to result in an absent or non-functional protein product. This variant is a well-known founder mutation in the Ashkenazi Jewish population and occurs at 0.96-1.14% minor allele frequency (PMID: 7550349, 8571953, 30152102). This variant has been reported in dozens of individuals and families affected with breast and ovarian cancer (PMID: 7894492, 7611277, 7837387, 8533757, 8531968, 8642955, 9042909, 9150153, 21643751, 30480775, 35020120). This variant also has been observed in individuals from diverse ethnicities in Africa, America, Asia and Europe (PMID: 8651293, 24312913). The risk of female breast cancer among carriers of this mutation is 55-83% by age 70, and the risk of ovarian cancer is 12-58% by age 70 (PMID: 9145676, 15994883, 22430266). A breast cancer case-control meta-analysis has detected this variant in 30/60436 cases and 8/53453 unaffected individuals with an odds ratio (OR) of 3.317 (95%CI 1.52 to 7.235) (PMID: 33471991; Leiden Open Variation Database DB-ID BRCA1_001114). This variant has been identified in 58/282442 chromosomes (42/10368 Ashkenazi Jewish chromosomes) in the general population by the Genome Aggregation Database (gnomAD). Loss of BRCA1 function is a known mechanism of disease (clinicalgenome.org). Based on the available evidence, this variant is classified as Pathogenic. ([less](#))

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown
Number of individuals with the variant: 78

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Zygosity: 78 Single Heterozygotes					
Pathogenic (May 09, 2016) N Not contributing to aggregate classification	★ ★ ★ ★ (EGL Classification Definitions 2015)	not provided	Eurofins Ntd Llc (ga) Accession: SCV000227400.6 First in ClinVar: Jun 28, 2015 Last updated: Apr 13, 2025		^
Publications: PubMed: 22430266					
Other databases http://lgdfm3.ncifcrf.gov/bic/BIC.html http://www.egl-eurofins.com/emvclass/emvclass.php?approved_symbol=BRCA1 http://www.umd.be/BRCA1/ https://portal.biobase-international.com/hgmd/pro/gene.php?gene=BRCA1					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: unknown Number of individuals with the variant: 4 Zygosity: 4 Single Heterozygotes Sex: mixed					
Pathogenic (Oct 23, 2020) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	not provided (Autosomal dominant inheritance)	Institute of Medical Genetics and Applied Genomics, University Hospital Tübingen Accession: SCV001447639.2 First in ClinVar: Nov 28, 2020 Last updated: Apr 13, 2025		^
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes Clinical Features: Breast carcinoma (present) Sex: female					
Pathogenic (Mar 05, 2022) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Dasa Accession: SCV002107151.3 First in ClinVar: Mar 28, 2022 Last updated: Apr 13, 2025		^
Publications: PubMed: 14576434, 21503673, 22430266, 7611277, 7894492					
Comment: The c.68_69del;p.(Glu23Valfs*17) is a null frameshift variant (NMD) in the BRCA1 gene and predicts alteration of the nonsense-mediate decay - NMD is present in a relevant exon to the transcript - PVS1. This sequence change has been observed in affected individual(s) and ClinVar contains an entry for this variant (ClinVar ID: 17662; PMID: 7611277; PMID: 7894492; PMID: 14576434; PMID: 21503673; PMID: 22430266)PS4. The variant is present at low allele frequencies population databases (rs80357914 – gnomAD 0.001446%; ABraOM no frequency - http://abraom.ib.usp.br/) - PM2_supporting. In summary, the currently available evidence indicates that the variant is pathogenic. (less)					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Sex: female Geographic origin: Brazil					
Pathogenic (Jun 17, 2022) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	MGZ Medical Genetics Center Accession: SCV002580991.2 First in ClinVar: Oct 15, 2022 Last updated: Apr 13, 2025 Comment: ACMG criteria applied: PVS1, PS4, PP1_MOD, PS3_SUP, PM2_SUP		^
Observation 1					
Collection method: clinical testing					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
	Allele origin: germline Affected status: yes Number of individuals with the variant: 3 Sex: female				
Pathogenic (Jun 08, 2023) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1 (Autosomal dominant inheritance)	Institute of Human Genetics, University of Leipzig Medical Center Accession: SCV004027730.4 First in ClinVar: Aug 26, 2023 Last updated: Apr 13, 2025		
Comment: Criteria applied: PVS1,PS4 (less)					
Observation 1					
Collection method: clinical testing Allele origin: unknown Affected status: yes Clinical Features: Ovarian carcinoma (present) Sex: female					
Pathogenic (Apr 02, 2024) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Familial cancer of breast (Autosomal dominant inheritance)	Institute of Human Genetics, University of Leipzig Medical Center Accession: SCV005368055.2 First in ClinVar: Oct 13, 2024 Last updated: Apr 13, 2025		
Comment: Criteria applied: PVS1,PS3,PS4_MOD,PM5_STR (less)					
Observation 1					
Collection method: clinical testing Allele origin: unknown Affected status: yes Clinical Features: Ovarian carcinoma (present) Sex: female					
Pathogenic (Oct 26, 2018) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Fanconi anemia, complementation group S	Department of Pathology and Laboratory Medicine, Sinai Health System Accession: SCV005915374.1 First in ClinVar: Apr 28, 2025 Last updated: Apr 28, 2025		
Publications: PubMed: 10353781, 10413426, 10573018, 10739756, 12677558, 12955716, 14576434, 14709740, 15994883, 18165636, 23055798, 7611277, 9634504					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes					
Pathogenic (Mar 31, 2025) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Hereditary cancer-predisposing syndrome	Color Diagnostics, LLC DBA Color Health Accession: SCV000292122.6 First in ClinVar: Jul 08, 2016 Last updated: May 03, 2025		
Publications: PubMed: 15994883, 21643751, 22430266, 24312913, 30152102, 30480775, 33471991, 35020120, 7550349, 7611277, 7837387, 7894492, 8531968, 8533757, 8571953, 8642955, 8651293, 9042909, 9145676, 9150153					
Comment: This variant deletes 2 nucleotides in exon 2 of the BRCA1 gene, causing a frameshift and a premature translational stop signal. This variant is also known as 185delAG and 187delAG in the literature. This variant is expected to result in an absent or non-functional protein product. This variant is a well-known founder mutation in the Ashkenazi Jewish population and occurs at 0.96-1.14% minor allele frequency (PMID: 7550349, 8571953, 30152102). This variant has been reported in dozens of individuals and families affected with breast and ovarian cancer (PMID: 7894492, 7611277, 7837387, 8533757, 8531968, 8642955, 9042909, 9150153, 21643751, 30480775, 35020120). This variant also has been observed in individuals from diverse ethnicities in Africa, America, Asia and Europe (PMID: 8651293, 24312913). The risk of female breast cancer among carriers of this mutation is 55-83% by age 70, and the risk of ovarian cancer is 12-58% by age 70 (PMID: 9145676, 15994883, 22430266). A breast cancer case-control meta-analysis has detected this variant in 30/60436 cases and 8/53453 unaffected individuals with an odds ratio (OR) of 3.317 (95%CI 1.52 to 7.235) (PMID: 33471991; Leiden Open Variation Database DB-ID BRCA1_001114). This variant has been identified in 58/282442 chromosomes (42/10368 Ashkenazi Jewish chromosomes) in the general population by the Genome Aggregation Database (gnomAD). Loss of BRCA1 function is a known mechanism of disease (clinicalgenome.org). Based on the available evidence, this variant is classified as Pathogenic. (less)					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: unknown Platform type: NGS					
Pathogenic (Sep 18, 2024) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1 (Autosomal dominant inheritance)	Molecular Pathology, Peter Maccallum Cancer Centre Accession: SCV006276411.1 First in ClinVar: Jul 13, 2025 Last updated: Jul 13, 2025	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: unknown	
Pathogenic (Jul 14, 2017) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1 (Autosomal dominant inheritance)	Bioinformatics dept., Datar Cancer Genetics Limited, India Accession: SCV000583605.3 First in ClinVar: Dec 06, 2016 Last updated: Aug 03, 2025		
Publications: PubMed: 23788959					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Indication for testing: Clinical features observed in the individual Zygosity: 1 Compound Heterozygote Age: 40-49 years Sex: female Ethnicity/Population group: South Asian Geographic origin: India Comment on evidence: "loss_of_function_variant" was previously submitted as the functional consequence for NM_007294.3:c.68_69delAG, but without providing the result of a functional assay.					
Pathogenic (Aug 14, 2025) N Not contributing to aggregate classification	★★★★ (CanVIG BRCA Gene Specific V1.22)	Inherited ovarian cancer (without breast cancer)	Genetics Laboratory, Great Ormond Street Hospital NHS Foundation Trust, North Thames Genomic Laboratory Hub Accession: SCV006560819.1 First in ClinVar: Oct 25, 2025 Last updated: Oct 25, 2025		
Comment: PS4_moderate, PVS1_very-strong, PS3_strong (less)					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes					
Pathogenic (Aug 01, 2024) N Not contributing to aggregate classification	★★★★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	3billion Accession: SCV006582731.1 First in ClinVar: Oct 25, 2025 Last updated: Oct 25, 2025		
Publications: PubMed: 7611277					
Comment: The variant is observed at an allele frequency greater than expected for the associated disorder in the gnomAD v4.0.0 dataset and therefore considered benign. Predicted Consequence/Location: Frameshift: predicted to result in a loss or disruption of normal protein function through nonsense-mediated decay (NMD) or protein truncation. Multiple pathogenic variants are reported downstream of the variant. The variant has been reported multiple times as an established pathogenic variant (ClinVar ID: VCV000017662 /PMID: 7611277 /3billion dataset). Therefore, this variant is classified as Pathogenic according to the recommendation of ACMG/AMP guideline. (less)					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes Method: exome sequencing					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Pathogenic (Dec 29, 2025) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	not provided	Center for Genomic Medicine, Rigshospitalet, Copenhagen University Hospital Accession: SCV002551076.6 First in ClinVar: Jul 23, 2022 Last updated: Jan 03, 2026		
Publications: PubMed: 8841191					
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown					
Pathogenic (Mar 08, 2024) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	not provided	Mayo Clinic Laboratories, Mayo Clinic Accession: SCV000778781.5 First in ClinVar: Jun 23, 2018 Last updated: Jan 11, 2026		
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown Number of individuals with the variant: 14					
Pathogenic (May 13, 2025) N Not contributing to aggregate classification	★ ★ ★ ★ (ACOS Best Practice Guidelines for Variant Classification in Rare Disease 2020)	Inherited breast cancer and ovarian cancer	Cambridge Genomics Laboratory, East Genomic Laboratory Hub, NHS Genomic Medicine Service Accession: SCV007299209.1 First in ClinVar: Jan 17, 2026 Last updated: Jan 17, 2026		
Comment: PVS1,PS4_Very Strong,PM5_Strong (less)					
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes					
Pathogenic (Jul 07, 2025) N Not contributing to aggregate classification	★ ★ ★ ★ (ARUP Molecular Germline Variant Investigation Process 2024)	not provided	ARUP Laboratories, Molecular Genetics and Genomics, ARUP Laboratories Accession: SCV000602669.8 First in ClinVar: Sep 30, 2017 Last updated: Jan 24, 2026		
Comment: The BRCA1 c.68_69delAG; p.Glu23ValfsTer17 variant (rs386833395), also known as 185delAG, is a well known founder variant associated with breast, ovarian, and pancreatic cancer in the Ashkenazi Jewish population as well as other ethnic populations (Abeliovich 1997, Antoniou 2005, King 2003, Laitman 2019, Lucas 2014). This variant is reported as pathogenic by multiple laboratories in ClinVar (Variation ID: 17662), and is found in the general population with an overall allele frequency of 0.02% (58/282442 alleles) in the Genome Aggregation Database. This variant causes a frameshift by deleting 2 nucleotides, so it is predicted to result in a truncated protein or mRNA subject to nonsense-mediated decay. Based on available information, the p.Glu23ValfsTer17 variant is considered to be pathogenic. References: Abeliovich D et al. The founder mutations 185delAG and 5382insC in BRCA1 and 6174delT in BRCA2 appear in 60% of ovarian cancer and 30% of early-onset breast cancer patients among Ashkenazi women. <i>Am J Hum Genet.</i> 1997 <i>Mar</i>;60(3):505-14. Antoniou AC et al. Breast and ovarian cancer risks to carriers of the BRCA1 5382insC and 185delAG and BRCA2 6174delT mutations: a combined analysis of 22 population based studies. <i>J Med Genet.</i> 2005 <i>Jul</i>;42(7):602-3. King MC et al. Breast and ovarian cancer risks due to inherited mutations in BRCA1 and BRCA2. <i>Science.</i> 2003 <i>Oct</i> 24;302(5645):643-6. Laitman Y et al. The spectrum of BRCA1 and BRCA2 pathogenic sequence variants in Middle Eastern, North African, and South European countries. <i>Hum Mutat.</i> 2019 <i>Nov</i>;40(11):e1-e23. Lucas AL et al. BRCA1 and BRCA2 germline mutations are frequently demonstrated in both high-risk pancreatic cancer screening and pancreatic cancer cohorts. <i>Cancer.</i> 2014 <i>Jul</i> 1;120(13):1960-7. (less)					
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Pathogenic (Dec 09, 2025) N Not contributing to aggregate classification	★ ★ ★ ★ (CanVIG BRCA Gene Specific V1.22)	Inherited breast cancer and ovarian cancer	Genetics Laboratory, Great Ormond Street Hospital NHS Foundation Trust, North Thames Genomic Laboratory Hub Accession: SCV007346608.1 First in ClinVar: Feb 15, 2026 Last updated: Feb 15, 2026		⬆
Comment: PS4_strong, PVS1_very-strong, PM5_strong (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes					
Pathogenic (-) N Not contributing to aggregate classification	★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer, familial, susceptibility to, 1	Genesis Genomics Accession: SCV007592300.1 First in ClinVar: May 09, 2026 Last updated: May 09, 2026		⬆
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Clinical Features: Breast cancer (present) Age: 30-39 years Sex: female					
Observation 2 Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Clinical Features: Breast cancer (present) Age: 60-69 years Sex: female					
Observation 3 Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Clinical Features: Breast cancer (present) Age: 30-39 years Sex: female					
Observation 4 Collection method: clinical testing Allele origin: germline Affected status: no Number of individuals with the variant: 1 Sex: female					
Observation 5 Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Clinical Features: Breast cancer (present) Age: 60-69 years Sex: female					
Pathogenic (Jul 24, 2014) N Not contributing to aggregate classification	★ ★ ★ ★	Breast-ovarian cancer, familial 1	Pathway Genomics Accession: SCV000189887.1 First in ClinVar: Oct 19, 2014 Last updated: Oct 19, 2014		⬆

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Publications: PubMed: 12473589, 15994883, 18034184, 20301425, 23233716, 24764757 Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown					
Pathogenic (Nov 14, 2013) N Not contributing to aggregate classification	★★★★	Breast-ovarian cancer, familial 1	Sharing Clinical Reports Project (SCRIP) Accession: SCV000053880.6 First in ClinVar: Apr 04, 2013 Last updated: Jun 24, 2016	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: not provided	
Pathogenic (Jan 31, 2014) N Not contributing to aggregate classification	★★★★	Hereditary breast ovarian cancer syndrome	Research Molecular Genetics Laboratory, Women's College Hospital, University of Toronto Study: The Canadian Open Genetics Repository (COGR) Accession: SCV000587006.1 First in ClinVar: Apr 16, 2017 Last updated: Apr 16, 2017	Observation: 1 - Collection method: research - Allele origin: germline - Affected status: yes	
Pathogenic (Oct 01, 2008) N Not contributing to aggregate classification	★★★★	BREAST-OVARIAN CANCER, FAMILIAL, SUSCEPTIBILITY TO, 1	OMIM Accession: SCV000039518.4 First in ClinVar: Apr 04, 2013 Last updated: Apr 09, 2018		

Publications:

PubMed: 12220453, 16941470, 18762988, 7550349, 7894492, 8841191, 9450187, 9536083, 9625172, 9764635

Observation 1

Collection method: literature only
Allele origin: germline
Affected status: not provided
Comment on evidence:
Breast-Ovarian Cancer Susceptibility Simard et al. (1994) studied 30 Canadian families with breast and/or ovarian cancer (604370) for germline mutations in the coding region of the BRCA1 candidate gene. They identified a 2-bp (AG185) deletion in the normal sequence TTA GAG of codons 22-23 in exon 3. The AGAG presumably predisposed to the deletion. This mutation changes the reading frame of the mRNA and causes a premature termination codon at position 39. This mutation was detected in index cases from 4 families that were not known to be related and originated from different areas in Canada. In these 4 families there were a total of 12 cases of breast cancer and 11 cases of ovarian cancer. Struewing et al. (1995) pointed out that all 10 published families with the 185delAG mutation (also called 187delAG) were Ashkenazi Jewish (of Eastern European origin). They knew of an eleventh Ashkenazi breast/ovarian cancer family with the 185delAG mutation; furthermore, only 1 Ashkenazi Jewish family was known to have a BRCA1 mutation other than 185delAG. In addition, Ashkenazi families with the 185delAG mutation appeared to share a common haplotype. In a study of 858 Ashkenazim seeking genetic testing for conditions unrelated to cancer, they observed the 185delAG mutation in 0.9% (95% confidence limit, 0.4%-1.8%), and in 815 reference individuals not selected for ethnic origin, none had the mutation. Roa et al. (1996) found the 185delAG mutation in 1.09% of approximately 3,000 Ashkenazi Jewish individuals and found the 5382insC mutation (113705.0018) in 0.13%. BRCA2 analysis on 3,085 individuals from the same population showed a carrier frequency of 1.52% for the 6174delT mutation (600185.0009). The expanded population-based study confirmed that the BRCA1 185delAG mutation and the BRCA2 6174delT mutation constituted the 2 most frequent mutant alleles predisposing to hereditary breast cancer among Ashkenazim and suggested a relatively lower penetrance for the 6174delT mutation in BRCA2. Bar-Sade et al. (1997) examined 639 unrelated healthy Jews of Iraqi extraction, a presumed low-risk group for the 185delAG mutation which occurs predominantly in Ashkenazim. Three individuals were identified as 185delAG mutation carriers, and haplotype analysis of the Iraqi mutation carriers showed that 2 of the Iraqis shared a haplotype in common with 6 Ashkenazi mutation carriers, and a third had a haplotype that differed by a single marker. This suggested to Bar-Sade et al. (1997) that the BRCA1 185delAG mutation may have arisen before the dispersion of the Jewish people in the Diaspora, at least at the time of Christ. Bar-Sade et al. (1998) extended their analyses to other non-Ashkenazi subsets: 354 of Moroccan origin, 200 Yemenites, and 150 Iranian Jews. Four of Moroccan origin (1.1%) and none of the Yemenites or Iranians were carriers of the 185delAG mutation. BRCA1 allelic patterns (haplotypes) were determined for 4 of these individuals and for 12 additional non-Ashkenazi 185delAG mutation carriers who had breast/ovarian cancer. The common 'Ashkenazi haplotype' was shared by 6 non-Ashkenazi individuals; 4 had a closely related pattern, and the rest (n = 6) displayed a distinct BRCA1 allelic pattern. The authors concluded that the 185delAG BRCA1 mutation occurs in some non-Ashkenazi populations at rates comparable with that of Ashkenazim. The majority of Jewish 185delAG mutation carriers have the same haplotype, supporting the founder effect notion, but dating the mutation's origin to an earlier date than previously estimated. The different allelic pattern at the BRCA1 locus in some Jewish mutation carriers might suggest that the mutation arose independently. Bandera et al. (1998) demonstrated the 185delAG mutation in 2 women with a personal or family history of breast cancer and papillary serous carcinoma of the peritoneum (PSCP). PSCP is histologically indistinguishable from serous epithelial ovarian carcinoma and it may develop years after oophorectomy. Schorge et al. (1998) demonstrated that the tumors were multifocal in these cases, indicating that patients with germline BRCA1 mutations may develop PSCP in addition to breast and ovarian carcinomas. Ah Mew et al. (2002) reported the 185delAG mutation in a non-Jewish Chilean family with no reported Jewish ancestry. The linked haplotype present in this family was identical to that identified in the Ashkenazi Jewish population. Buisson et al. (2006) found that BRCA1 transcripts bearing the 185delAG mutation are not degraded by nonsense-mediated mRNA decay. Using Western blot analysis, they examined HeLa cells transfected with minigenes for this transcript and another with a premature termination codon at position 36 and found that translation from these transcripts was reinitiated at codon 128. Pancreatic Cancer Susceptibility Al-Sukhni et al. (2008) found loss of heterozygosity at the BRCA1 locus in pancreatic tumor DNA from 5 (71%) of 7 patients with pancreatic cancer (PNCA4; 614320) who carried a heterozygous germline BRCA1 mutation. Three patients carried the 185delAG mutation. In contrast, only 1 (11%) of 9 patients with sporadic pancreatic cancer and no germline BRCA1 mutations showed LOH at the BRCA1 locus. Al-Sukhni et al. (2008) concluded that BRCA1 germline mutations likely predispose to the development of pancreatic cancer, and suggested that individuals with these mutations be considered for pancreatic cancer-screening programs. (less)

risk factor (Oct 01, 2008) N Not contributing to aggregate classification	★★★★	PANCREATIC CANCER, SUSCEPTIBILITY TO, 4	OMIM Accession: SCV000053475.4 First in ClinVar: Apr 04, 2013 Last updated: Dec 15, 2018	
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Publications:

PubMed: 12220453, 16941470, 18762988, 7550349, 7894492, 8841191, 9450187, 9536083, 9625172, 9764635

Observation 1

Collection method: literature only
Allele origin: germline
Affected status: not provided

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Comment on evidence: Breast-Ovarian Cancer Susceptibility Simard et al. (1994) studied 30 Canadian families with breast and/or ovarian cancer (604370) for germline mutations in the coding region of the BRCA1 candidate gene. They identified a 2-bp (AG185) deletion in the normal sequence TTA GAG of codons 22-23 in exon 3. The AGAG presumably predisposed to the deletion. This mutation changes the reading frame of the mRNA and causes a premature termination codon at position 39. This mutation was detected in index cases from 4 families that were not known to be related and originated from different areas in Canada. In these 4 families there were a total of 12 cases of breast cancer and 11 cases of ovarian cancer. Struewing et al. (1995) pointed out that all 10 published families with the 185delAG mutation (also called 187delAG) were Ashkenazi Jewish (of Eastern European origin). They knew of an eleventh Ashkenazi breast/ovarian cancer family with the 185delAG mutation; furthermore, only 1 Ashkenazi Jewish family was known to have a BRCA1 mutation other than 185delAG. In addition, Ashkenazi families with the 185delAG mutation appeared to share a common haplotype. In a study of 858 Ashkenazim seeking genetic testing for conditions unrelated to cancer, they observed the 185delAG mutation in 0.9% (95% confidence limit, 0.4%-1.8%), and in 815 reference individuals not selected for ethnic origin, none had the mutation. Roa et al. (1996) found the 185delAG mutation in 1.09% of approximately 3,000 Ashkenazi Jewish individuals and found the 5382insC mutation (113705.0018) in 0.13%. BRCA2 analysis on 3,085 individuals from the same population showed a carrier frequency of 1.52% for the 6174delT mutation (600185.0009). The expanded population-based study confirmed that the BRCA1 185delAG mutation and the BRCA2 6174delT mutation constituted the 2 most frequent mutant alleles predisposing to hereditary breast cancer among Ashkenazim and suggested a relatively lower penetrance for the 6174delT mutation in BRCA2. Bar-Sade et al. (1997) examined 639 unrelated healthy Jews of Iraqi extraction, a presumed low-risk group for the 185delAG mutation which occurs predominantly in Ashkenazim. Three individuals were identified as 185delAG mutation carriers, and haplotype analysis of the Iraqi mutation carriers showed that 2 of the Iraqis shared a haplotype in common with 6 Ashkenazi mutation carriers, and a third had a haplotype that differed by a single marker. This suggested to Bar-Sade et al. (1997) that the BRCA1 185delAG mutation may have arisen before the dispersion of the Jewish people in the Diaspora, at least at the time of Christ. Bar-Sade et al. (1998) extended their analyses to other non-Ashkenazi subsets: 354 of Moroccan origin, 200 Yemenites, and 150 Iranian Jews. Four of Moroccan origin (1.1%) and none of the Yemenites or Iranians were carriers of the 185delAG mutation. BRCA1 allelic patterns (haplotypes) were determined for 4 of these individuals and for 12 additional non-Ashkenazi 185delAG mutation carriers who had breast/ovarian cancer. The common 'Ashkenazi haplotype' was shared by 6 non-Ashkenazi individuals; 4 had a closely related pattern, and the rest (n = 6) displayed a distinct BRCA1 allelic pattern. The authors concluded that the 185delAG BRCA1 mutation occurs in some non-Ashkenazi populations at rates comparable with that of Ashkenazim. The majority of Jewish 185delAG mutation carriers have the same haplotype, supporting the founder effect notion, but dating the mutation's origin to an earlier date than previously estimated. 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Using Western blot analysis, they examined HeLa cells transfected with minigenes for this transcript and another with a premature termination codon at position 36 and found that translation from these transcripts was reinitiated at codon 128. Pancreatic Cancer Susceptibility Al-Sukhni et al. (2008) found loss of heterozygosity at the BRCA1 locus in pancreatic tumor DNA from 5 (71%) of 7 patients with pancreatic cancer (PNCA4; 614320) who carried a heterozygous germline BRCA1 mutation. Three patients carried the 185delAG mutation. In contrast, only 1 (11%) of 9 patients with sporadic pancreatic cancer and no germline BRCA1 mutations showed LOH at the BRCA1 locus. Al-Sukhni et al. (2008) concluded that BRCA1 germline mutations likely predispose to the development of pancreatic cancer, and suggested that individuals with these mutations be considered for pancreatic cancer-screening programs. (less)					
Pathogenic (Dec 01, 2018) N Not contributing to aggregate classification	★★★★	Ovarian neoplasm	German Consortium for Hereditary Breast and Ovarian Cancer, University Hospital Cologne Accession: SCV000923765.1 First in ClinVar: Jun 17, 2019 Last updated: Jun 17, 2019		
Observation 1 Collection method: research Allele origin: germline Affected status: yes Platform type: next-gen sequencing					
Pathogenic (-) N Not contributing to aggregate classification	★★★★	not provided	Joint Genome Diagnostic Labs from Nijmegen and Maastricht, Radboudumc and MUMC+ Study: VKGL Data-share Consensus Accession: SCV002037545.1 First in ClinVar: Dec 25, 2021 Last updated: Dec 25, 2021	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: yes	
Pathogenic (Feb 16, 2024) N Not contributing to aggregate classification	★★★★	BRCA1-related condition	PreventionGenetics, part of Exact Sciences Accession: SCV000806982.4 First in ClinVar: Jun 23, 2018 Last updated: Oct 08, 2024		
Comment: The BRCA1 c.68_69delAG variant is predicted to result in a frameshift and premature protein termination (p.Glu23Valfs*17). This variant, also described as 185delAG or 187delAG in the literature, has been associated with autosomal dominant hereditary breast and ovarian cancer syndrome (HBOC) (OMIM #604370). This is a recurrent variant among individuals of Spanish origin and is a founder variant in the Ashkenazi Jewish population, identified in about 1% of Ashkenazi Jewish individuals unselected for breast cancer (Gabalδό Barrios et al. 2017. PubMed ID: 28477318; Finkelman et al. 2012. PubMed ID: 22430266; Struewing et al. 1997. PubMed ID:9145676; Roa et al. 1996. PubMed ID:8841191). This variant is frequently observed in the Ashkenazi Jewish population in gnomad. This variant is also found in individuals of diverse ancestry at variable frequencies. It is interpreted as pathogenic in ClinVar (https://www.ncbi.nlm.nih.gov/clinvar/variation/17662/). In summary, this variant is interpreted as pathogenic. (less)					
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown					
Pathogenic (Jul 13, 2012) N Not contributing to aggregate classification	★★★★	Breast-ovarian cancer, familial	Biesecker Lab/Clinical Genomics Section, National Institutes of Health Study: ClinSeq Accession: SCV000043189.3 First in ClinVar: Apr 12, 2013 Last updated: Jun 08, 2025		
Publications: PubMed: 22703879					
Comment: Converted during submission from pathogenic to Pathogenic. (less)					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Observation 1					
Collection method: research Allele origin: germline Affected status: no Number of individuals with the variant: 1 Zygosity: 1 Single Heterozygote Age: 60-69 years Sex: male Ethnicity/Population group: Ashkenazi Jewish Comment on evidence: The study set was not selected for affection status in relation to any cancer. Pathogenicity categories were based on literature curation. See Pubmed ID:22703879 for details. (less) Platform type: High-Throughput DNA Sequencing Platform name: Illumina GAlIx					
Pathogenic (May 29, 2002)	★★★★	Breast-ovarian cancer, familial 1	Breast Cancer Information Core (BIC) (BRCA1) Accession: SCV000144169.2 First in ClinVar: Apr 01, 2014 Last updated: Oct 11, 2015		
N Not contributing to aggregate classification Publications: PubMed: 15024741, 15383404 Family is not Ashkenazi Jewish. Strewing et al 1995					
Observation 1					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 599					
Observation 2					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 32 Geographic origin: Ashkenazi					
Observation 3					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Geographic origin: Ashkenazi, Central Eastern European					
Observation 4					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Geographic origin: Ashkenazi, Central/Eastern European					
Observation 5					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 2 Geographic origin: Austria					
Observation 6					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Geographic origin: Central/Eastern European					
Observation 7					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 7 Geographic origin: Netherlands					
Observation 8					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Geographic origin: France					
Observation 9					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Geographic origin: Hispanic					
Observation 10					
Collection method: clinical testing					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
	Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Geographic origin: Mexican American				
	Observation 11				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Geographic origin: Native American				
	Observation 12				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Geographic origin: Western European, Ashkenazi				
	Observation 13				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: African				
	Observation 14				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: African, Latin American, Caribbean				
	Observation 15				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1075 Ethnicity/Population group: Ashkenazi				
	Observation 16				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 3 Ethnicity/Population group: Ashkenazi Geographic origin: France				
	Observation 17				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 6 Ethnicity/Population group: Ashkenazi Jewish				
	Observation 18				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi Jewish Geographic origin: American				
	Observation 19				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi Jewish Geographic origin: Canada				
	Observation 20				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi Jewish Geographic origin: Germany				
	Observation 21				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 31 Ethnicity/Population group: Ashkenazi, Central/Eastern European				
	Observation 22				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi, Central/Eastern European, Nea				
	Observation 23				
	Collection method: clinical testing Allele origin: germline Affected status: yes				

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
	Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi, Dutch, Italian				
	Observation 24				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi, Englishraine				
	Observation 25				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi, Greek				
	Observation 26				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi, Scandinavian				
	Observation 27				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 2 Ethnicity/Population group: Ashkenazi, Sephardic				
	Observation 28				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi, Sephardic Jewish				
	Observation 29				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 5 Ethnicity/Population group: Ashkenazi, Western European				
	Observation 30				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Ashkenazi, Western, Central/Eastern European				
	Observation 31				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 2 Ethnicity/Population group: Asian				
	Observation 32				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Asian, Ashkenazi, Western European				
	Observation 33				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 2 Ethnicity/Population group: Asian, Indian				
	Observation 34				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Austrian, Jewish, Polish				
	Observation 35				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Caucasian				
	Observation 36				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 6 Ethnicity/Population group: Caucasian Geographic origin: Czech Republic				

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	^
Observation 37					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 6 Ethnicity/Population group: Caucasian Geographic origin: Netherlands					
Observation 38					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 5 Ethnicity/Population group: Caucasian Geographic origin: English					
Observation 39					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Caucasian Geographic origin: Italy					
Observation 40					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 6 Ethnicity/Population group: Caucasian Geographic origin: Spain					
Observation 41					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 7 Ethnicity/Population group: Caucasian Non Hispanic					
Observation 42					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Caucasian, Non Ashkenazi					
Observation 43					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 17 Ethnicity/Population group: Central/Eastern European					
Observation 44					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Central/Eastern European, Russian					
Observation 45					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Central/Eastern European, Russian, Polish					
Observation 46					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Central/Eastern European, Turkish, Greek					
Observation 47					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Chilean					
Observation 48					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: English					
Observation 49					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: English, Jewish					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Observation 50					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Englishraine German					
Observation 51					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 2 Ethnicity/Population group: Gypsy Geographic origin: Spain					
Observation 52					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 3 Ethnicity/Population group: Indian Geographic origin: Malaysia					
Observation 53					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Irish, German, Scottish, English, Welsh					
Observation 54					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 54 Ethnicity/Population group: Jewish					
Observation 55					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Jewish Geographic origin: Chile					
Observation 56					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Jewish Geographic origin: Germany					
Observation 57					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Jewish Geographic origin: Norway					
Observation 58					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Jewish, Mixed					
Observation 59					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 21 Ethnicity/Population group: Latin American, Caribbean					
Observation 60					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Latin American, Caribbean, New Mexican					
Observation 61					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Latvian Geographic origin: Latvia					
Observation 62					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Mediterranean					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Observation 63					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 2 Ethnicity/Population group: Mid, Near East					
Observation 64					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 6 Ethnicity/Population group: Native American					
Observation 65					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Native American, Western European					
Observation 66					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 5 Ethnicity/Population group: Near Eastern					
Observation 67					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Near Eastern, Sephardic Jewish					
Observation 68					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Patient Not Ashkenazi Jewish					
Observation 69					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Romanian, Ashkenazi Jewish					
Observation 70					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Romanian, Jewish					
Observation 71					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Russian Jewish					
Observation 72					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Russian, Ashkenazi Jewish					
Observation 73					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Russian, Polish					
Observation 74					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 2 Ethnicity/Population group: Spanish					
Observation 75					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 87 Ethnicity/Population group: Western European					
Observation 76					
Collection method: clinical testing Allele origin: germline					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
	Affected status: yes Number of individuals with the variant: 23 Ethnicity/Population group: Western European, Ashkenazi				
	Observation 77				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Ashkenazi, Central, Eas				
	Observation 78				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Ashkenazi, Central/Eastern European, Polish, Austrian				
	Observation 79				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Ashkenazi, English, Dutch				
	Observation 80				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Central/Eastern European				
	Observation 81				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Danish, German, Neth				
	Observation 82				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, French				
	Observation 83				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, French Canadian				
	Observation 84				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, French, English, Irish				
	Observation 85				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 2 Ethnicity/Population group: Western European, French, Italian				
	Observation 86				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, French, Spanish				
	Observation 87				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Jewish				
	Observation 88				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 2 Ethnicity/Population group: Western European, Latin American, Caribbe				
	Observation 89				
	Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Native American, Centr				

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	^
Observation 90					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Native American, Latin					
Observation 91					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Near Eastern Mid Eastern					
Observation 92					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Spanish Mexican					
Observation 93					
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Western European, Central/Eastern European					
Observation 94					
Collection method: clinical testing Allele origin: unknown Affected status: yes Number of individuals with the variant: 6 Ethnicity/Population group: Ashkenazi Geographic origin: American					
Observation 95					
Collection method: clinical testing Allele origin: unknown Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Caucasian Geographic origin: American					

Pathogenic (N Not contributing to aggregate classification	★★★★	Breast-ovarian cancer, familial, susceptibility to, 1	Diagnostic Laboratory, Department of Genetics, University Medical Center Groningen Study: VKGL Data-share Consensus Accession: SCV000733684.1 First in ClinVar: Apr 09, 2018 Last updated: Apr 09, 2018	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: yes
Pathogenic (Jun 22, 2016) N Not contributing to aggregate classification	★★★★	Breast and/or ovarian cancer	Foulkes Cancer Genetics LDI, Lady Davis Institute for Medical Research Study: The Canadian Open Genetics Repository (COGR) Accession: SCV000863618.1 First in ClinVar: Dec 24, 2018 Last updated: Dec 24, 2018	Observation: 1 • Collection method: clinical testing • Allele origin: germline • Affected status: unknown
Pathogenic (Jun 11, 2019) N Not contributing to aggregate classification	★★★★	Breast and/or ovarian cancer	CZECANCA consortium Accession: SCV001451786.1 First in ClinVar: Dec 26, 2020 Last updated: Dec 26, 2020	^
Publications: PubMed: 32295079				
Observation 1				
Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 6 Ethnicity/Population group: Slavic Geographic origin: Czech Republic				
Pathogenic (N Not contributing to aggregate classification	★★★★	Malignant tumor of breast	Department of Pathology and Laboratory Medicine, Sinai Health System Additional submitter: Franklin by Genoox Study: The Canadian Open Genetics Repository (COGR) Accession: SCV000591232.2	^

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
First in ClinVar: Aug 27, 2017 Last updated: Apr 13, 2021					
Comment: The p.Glu23ValfsX17 deletion variant is one of three known founder pathogenic mutations common to individuals of Ashkenazi Jewish descent (Petrucelli 1998). This deletion is predicted to cause a frameshift, which alters the protein's amino acid sequence beginning at codon 23 and leads to a premature stop codon 17 codons downstream. This alteration is then predicted to lead to a truncated or absent protein product and loss of function. Loss of function of the BRCA1 gene is an established disease mechanism in hereditary breast and ovarian cancer. In summary, based on the above information, this variant is classified as pathogenic. (less)					
Observation 1 Collection method: clinical testing Allele origin: unknown Affected status: yes Number of individuals with the variant: 28					
Pathogenic (-) N Not contributing to aggregate classification	★★★★	not provided	Clinical Genetics Laboratory, Department of Pathology, Netherlands Cancer Institute Study: VKGL Data-share Consensus Accession: SCV001905815.1 First in ClinVar: Sep 23, 2021 Last updated: Sep 23, 2021	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: yes	
Pathogenic (-) N Not contributing to aggregate classification	★★★★	not provided	Clinical Genetics DNA and cytogenetics Diagnostics Lab, Erasmus MC, Erasmus Medical Center Study: VKGL Data-share Consensus Accession: SCV001965465.1 First in ClinVar: Oct 07, 2021 Last updated: Oct 07, 2021	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: yes	
Pathogenic (Aug 26, 2022) N Not contributing to aggregate classification	★★★★	Breast-ovarian cancer, familial, susceptibility to, 1	BRCAlab, Lund University Accession: SCV002589061.1 First in ClinVar: Apr 01, 2023 Last updated: Apr 01, 2023	Observation: 1 - Collection method: clinical testing - Allele origin: germline - Affected status: yes	
Pathogenic (Feb 21, 2023) N Not contributing to aggregate classification	★★★★	Endometrial carcinoma	CZECANCA consortium Accession: SCV003804349.1 First in ClinVar: Feb 25, 2023 Last updated: Feb 25, 2023		
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes Number of individuals with the variant: 1 Ethnicity/Population group: Slavic Geographic origin: Czech Republic					
Pathogenic (Mar 11, 2023) N Not contributing to aggregate classification	★★★★	Triple-negative breast cancer (Autosomal dominant inheritance)	Cancer Genomics Lab, PINUM Cancer Hospital Accession: SCV003914825.1 First in ClinVar: May 06, 2023 Last updated: May 06, 2023		
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes Clinical Features: Multifocal breast carcinoma (present) Zygosity: 1 Single Heterozygote Age: 40-49 years Ethnicity/Population group: Punjabi Geographic origin: Punjab Platform type: Next-generation sequencing Platform name: Ion GeneStudio S5 System					
Pathogenic (May 24, 2021) N Not contributing to aggregate classification	★★★★	Breast-ovarian cancer, familial, susceptibility to, 1	Molecular Oncology, Hospital Universitario Central de Asturias (HUCA) Accession: SCV005061273.1 First in ClinVar: Oct 13, 2024 Last updated: Oct 13, 2024		

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Publications: PubMed: 38922859 Observation 1 Collection method: case-control Allele origin: germline Affected status: yes Secondary finding: no					
Pathogenic (-) N Not contributing to aggregate classification	★★★★	Breast and ovarian cancer (Autosomal dominant inheritance)	Kasturba Medical College, Manipal, Kasturba Medical College, Manipal, Manipal Academy of Higher Education, Manipal, India Accession: SCV002032172.2 First in ClinVar: Dec 18, 2021 Last updated: Apr 13, 2025		
Observation 1 Collection method: clinical testing Allele origin: unknown Affected status: yes Age: 30-39 years Sex: female					
Pathogenic (May 05, 2023) N Not contributing to aggregate classification	★★★★	Breast-ovarian cancer, familial, susceptibility to, 1	KCCC/NGS Laboratory, Kuwait Cancer Control Center Accession: SCV003927207.2 First in ClinVar: Jun 03, 2023 Last updated: Apr 13, 2025		
Publications: PubMed: 20104584 Comment: This sequence change deletes 2 nucleotides from exon 2 of the BRCA1 mRNA (c.68_69delAG), causing a frameshift after codon 23 and the creation of a premature translation stop signal 17 amino acid residues later – p.Glu23Valfs*17). This is expected to result in an absent or disrupted protein product. This variant, also known as BRCA1_185delAG or 187delAG, is one of three main pathogenic founder variants in the Ashkenazi Jewish population although it has been observed in individuals from other ethnicities as well (PMID: 23199084). This mutation has been described in the mutation database ClinVar (Variation ID: 17662). Therefore, this variant has been classified as pathogenic. (less) Observation 1 Collection method: clinical testing Allele origin: germline Affected status: yes Sex: female					
Pathogenic (May 30, 2014) N Not contributing to aggregate classification	★★★★	Breast-ovarian cancer, familial 1 (Autosomal dominant inheritance)	Counsyl Accession: SCV000220360.3 First in ClinVar: Mar 29, 2015 Last updated: Jul 05, 2025		
Publications: PubMed: 11352856, 17307836, 21324516, 7550349, 8531968 Comment: This submission and the accompanying classification are no longer maintained by the submitter. For more information on current observations and classification, please contact variantquestions@myriad.com. (less) Observation 1 Collection method: literature only Allele origin: unknown Affected status: unknown					
not provided (-) N Not contributing to aggregate classification	★★★★	Hereditary breast ovarian cancer syndrome	GeneReviews Accession: SCV000086950.3 First in ClinVar: Oct 01, 2013 Last updated: Oct 01, 2022		
Publications:					

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
	PubMed: 11466700 , 11896095 , 12402332 , 17591843 , 30152102 BookShelf: NBK1247 Comment: Founder variant in Ashkenazi Jews; accounts for 72% of pathogenic variants in this population (less)				
Observation 1					
Collection method: literature only Allele origin: germline Affected status: unknown					

not provided (⁻) N Not contributing to aggregate classification	★★★★	Hereditary breast ovarian cancer syndrome	GenomeConnect - Invitae Patient Insights Network Accession: SCV001749327.3 First in ClinVar: Jul 18, 2021 Last updated: Apr 13, 2025	
Comment: Variant reported in multiple GenomeConnect-Invitae Patient Insights Network participants by Lab Invitae. Variant interpreted as Pathogenic and reported, most recently, on 02-06-2021 and on 03-26-2018. GenomeConnect-Invitae Patient Insights Network assertions are reported exactly as they appear on the patient-provided report from the testing laboratory. Registry team members make no attempt to reinterpret the clinical significance of the variant. Phenotypic details are available under supporting information. (less)				
Observation 1				
Collection method: phenotyping only Allele origin: unknown Affected status: unknown Clinical Features: Bladder neoplasm (present) Indication for testing: Diagnostic Zygosity: 1 Single Heterozygote Age: 80 years Sex: male Method: Gene Panel Sequencing Testing laboratory: Labcorp Genetics (formerly Invitae), Labcorp Date variant was reported to submitter: 2021-02-06 Testing laboratory interpretation: Pathogenic				
Observation 2				
Collection method: phenotyping only Allele origin: unknown Affected status: unknown Clinical Features: Family history of cancer (present) Indication for testing: Diagnostic Zygosity: 1 Single Heterozygote Age: 60-69 years Sex: male Method: Gene Panel Sequencing Testing laboratory: Labcorp Genetics (formerly Invitae), Labcorp Date variant was reported to submitter: 2018-03-26 Testing laboratory interpretation: Pathogenic				

not provided (⁻) N Not contributing to aggregate classification	★★★★	Breast-ovarian cancer, familial, susceptibility to, 1	CHARM Consortium Accession: SCV007127687.1 First in ClinVar: Jan 03, 2026 Last updated: Jan 03, 2026	
Observation 1				
Collection method: phenotyping only Allele origin: germline Affected status: yes Clinical Features: Breast carcinoma (present) , Ovarian neoplasm (present) Sex: female Tissue: blood Comment on evidence: Observed in 1 female with a germline BRCA1 variant in c. 185delAG; p.Glu23fs*17 (also commonly known as c.68_69del; p.Glu23fs*17), with a history of breast cancer at 47 years of age, ovarian cancer at 58 years of age and ovarian recurrence at 63 years of age. (less) Testing laboratory: unknown				

Citations for germline classification of this variant 

Title	Author	Journal	Year	Link
BRCA1- and BRCA2-Associated Hereditary Breast and Ovarian Cancer.	Adam MP <i>et al.</i>	-	2026	PMID: 20301425
Featuring BRCA1 and BRCA2 germline mutational landscape from Asturias (North Spain).	Pitiot AS <i>et al.</i>	Clinical genetics	2024	PMID: 38922859
Spectrum of germline pathogenic variants using a targeted next generation sequencing panel and genotype-phenotype correlations in patients with suspected hereditary breast cancer at an academic medical centre in Pakistan.	Akbar F <i>et al.</i>	Hereditary cancer in clinical practice	2022	PMID: 35710434
Pathogenic BRCA1 variants disrupt PLK1-regulation of mitotic spindle orientation.	He Z <i>et al.</i>	Nature communications	2022	PMID: 35459234
Chasing the origin of 23 recurrent BRCA1 mutations in Pakistani breast and ovarian cancer patients.	Rashid MU <i>et al.</i>	International journal of cancer	2022	PMID: 35377489
Germline Mutation Analysis in Sporadic Breast Cancer Cases With Clinical Correlations.	Ajaz S <i>et al.</i>	Frontiers in genetics	2022	PMID: 35356428
Detection of germline variants in Brazilian breast cancer patients using multigene panel testing.	Guindalini RSC <i>et al.</i>	Scientific reports	2022	PMID: 35264596
Value of the loss of heterozygosity to BRCA1 variant classification.	Santana Dos Santos E <i>et al.</i>	NPJ breast cancer	2022	PMID: 35039532
Analysis of pathogenic variants in BRCA1 and BRCA2 genes using next-generation sequencing in women with triple negative breast cancer from South India.	Rajagopal T <i>et al.</i>	Molecular biology reports	2022	PMID: 35020120
High likelihood of actionable pathogenic variant detection in breast cancer genes in women with very early onset breast cancer.	Evans DG <i>et al.</i>	Journal of medical genetics	2022	PMID: 33758026

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